APM2000 Datasheet

Particulate Matter Sensor

- Laser scattering principle, minimum particle size 0.3μm
- Unique laser self-calibration technology
- Standard digital output
- Full metal shielding shell, stronger anti-interference performance
- small size 49×31.5×10.8mm

Product Summary

APM2000 is a digital particle detection sensor designed based on the principle of laser scattering, which can detect particle concentration in real time. APM2000 can test the particle size range of 0.3 to 10μm, provides a variety of different digital output interfaces, and has the function of self-calibration when it is turned on. Has good stability; small size, easy to integrate.

Applications

APM2000 has a wide range of application scenarios and is suitable for air purifiers, fresh air systems, air quality monitoring equipment, air conditioners, portable instruments and other equipment.

Figure 1. APM2000

1 Working Principle

APM2000 contains a laser light source and a photodetector, which collects the scattered light intensity in real time. When the air containing particles flows through the air duct of the sensor, the particles in the air in the air duct cause laser scattering, and the photoelectric element collects the change of the scattered light intensity. The subsequent acquisition circuit calculates the equivalent particle size of the particles based on the MIE theory. diameter and the number of particles of different particle sizes per unit volume. The working principle of the sensor is shown in Figure 2.

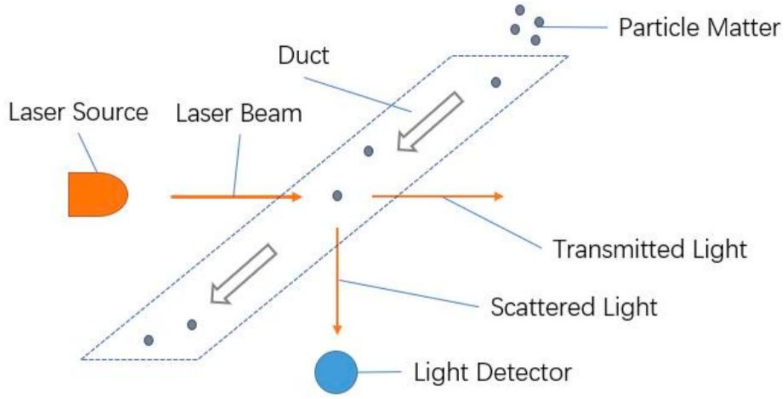


Figure 2. Sensor working principle

2 Sensor Characteristic

Table 1. Sensor characteristic

Parameter	Minimum	Typical	Maximum	Unit
Supply voltage	4.75	5	5.25	V DC
Operating current	-	50	100	mA
Standby current	-	10	-	mA
Particle size	0.3	2.5	10	µm
Particle concentration	0	-	1000	µg/m³
Accuracy of PM2.5[1]	±15µg/m³(0~100µg/m³) ±15%M.V(100~1000µg/m³)			-
Data update period	1			s
Stable time after power on	60			s
Lifetime[2]	>3			Y

[1] Test condition: Temperature of 25±2℃; humidity of 50±10%RH; reference instrument is TSI8530; source of particle matter is from cigarettes.

[2] Depending on the operating environment.

Dimension	47×37×12.4			mm
Working temperature	-10	25	50	℃
Storage temperature	-30	25	70	℃

3 Interface and Communication Protocol

3.1 APM2000 Pin

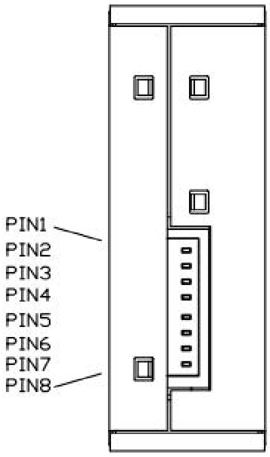


Figure 3. APM2000 pin

Table 2. Pin definition

	Name	Function
Pin 1	VCC	Power
Pin 2	GND	Ground
Pin 3	SET	0-I^2C; 1 or float-UART
Pin 4	RX/SDA	UART receive/I^2C clock
Pin 5	TX/SCL	UART send/I^2C clock
Pin 6	NC	-
Pin 7	NC	-
Pin 8	PWM	PWM

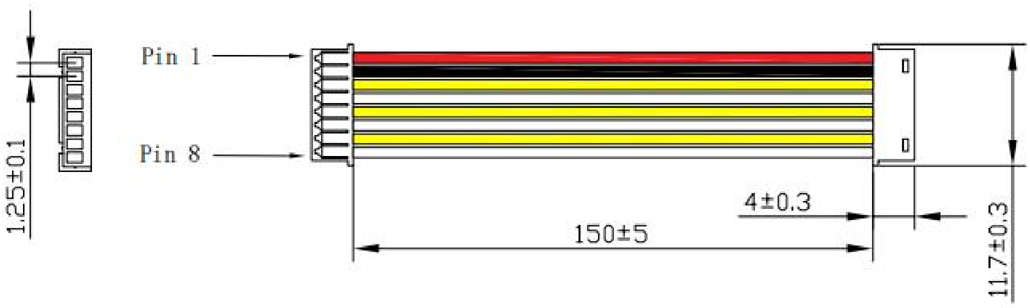


Figure 4. APM2000 cable (unit:mm; connector: MX1.25-8Y)

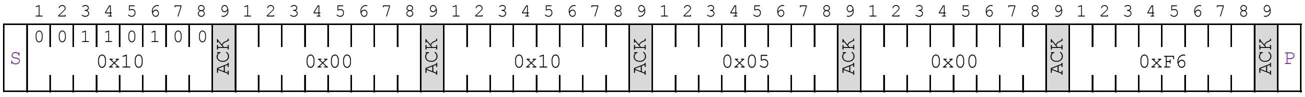
3.2 Communication interface

APM2000 supports the standard I^2C communication protocol, including two pins of SDA and SCL, and the two pins need to be externally connected with 2-10 pull-up resistors to VCC. The I^2C communication address of APM2000 is 0x08 (7-bit), the write command is 0x10, and the read command is 0x11.

3.3 I^2C communication

3.3.1 Start measurement: 10 00 10 05 00 F6

Send this command to enter measurement mode.

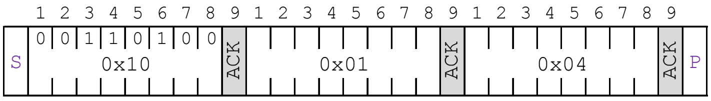


master signal slave signal S: start P: stop ACK: response NACK: non-response

In the following commands, the signal meanings are explained the same.

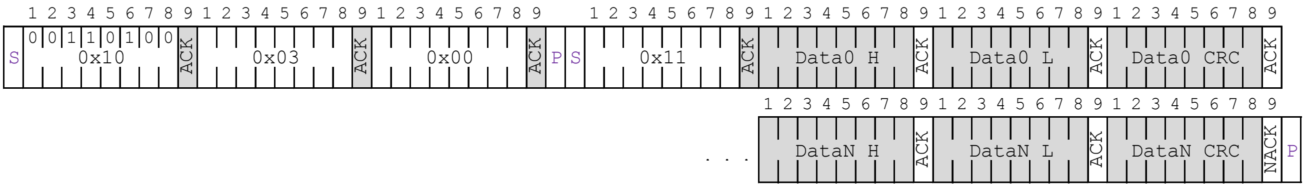
3.3.2 Stop measurement: 10 01 04

Send this command to exit the measurement mode.



3.3.3 read value: 10 03 00 11 Data0_H Data0_L Data0_CRC ......

Send this command to read the measured value.



Read 30 bytes of valid data, the data format is as follows:

Table 3: Data description

Byte	Data	Description
0~2	The measured value of each particulate matter is 16-bit data, the high byte comes first, the low byte follows, and two bytes are followed by a CRC check value	PM1.0 concentration=byte 0×256+byte 1(unit:μg/m^3); byte 2 is byte 0, byte 1 is CRC check
3~5		PM2.5 concentration=byte3×256+byte4(unit:μg/m^3); byte5 is byte3 and byte4 CRC check
6~8		Reserved
9~11		PM10 concentration=byte9×256+byte10 (unit:μg/m^3); byte11is byte9 and byte10 CRC check
12~14		Reserved humidity=byte12×256 + byte13 (unit:%RH, *10times); byte14 is byte12 and byte13 CRC check

15~17		Reserved temperature=byte15×256 + byte16 (unit: ℃, *10 times); Less than 500 is a negative temperature value, such as: 400 is -10°C, 600 is 10°C, Byte 17 is the CRC check value of byte 15 and byte 16
18~20		Reserved
21~23		Reserved
24~26		Reserved
27~29		Reserved

The sensor adopts CRC8 check, the initial value is 0xFF, and the polynomial is 0x31($x^8+x^5+x^4+1$), code as below:

```
//*****************************************************************
//function name:Calc_CRC8
//Function: CRC8 calculation, Initial value:0xFF, polynomial:
0x31(x8 + x5 + x4 + 1)
//Parameter: u8*dat: The first address of the data to be verified;
u8 Num: CRC verification data length
//Return: crc: Calculated Check value
//*****************************************************************
unsigned char Calc_CRC8(unsigned char  *data, unsigned char  Num)
{
    unsigned char bit,byte,crc=0xFF;

    for(byte=0; byte<Num; byte++)
    {
        crc^=(data[byte]);
        for(bit=8;bit>0;--bit)
        {
            if(crc&0x80)
                crc=(crc<<1)^0x31;
            else
                crc=(crc<<1);
        }
    }
    return crc;
```

3.4 UART communication protocol

3.4.1 Serial communication settings

Table 4. Serial communication parameters

Data	Parameter
Baud rate	1200
Data byte	8 bytes
Stop byte	1 byte
Check	no

3.4.2 UART

Table 5. UART protocol format

Frame header	Fixed code	Length(1 byte)	Command(1 byte)	Data(n byte)	check sum
FE	A5	XX	XX	XX	CS

Table 6. UART protocol description

Protocol domain	Description
Frame header	Value fixed to FE
Fixed code	Sensor category, value fixed to A5
Length	Length of data
Command	Operation instruction code
Data	Data to be read or written
Check sum	Data cumulative sum (lower 8 bits) = fixed code + length + command code + data

3.4.3 Serial port protocol command code table

Table 7. Serial port protocol command code table

Function	Command
Read PM2.5 concentration	0x00
Read PM1.0, PM2.5, PM10 concentration	0x01
Read PM2.5, humidity, temperature result	0x02(reserved)

a)Read PM2.5 concentration

Table 8. Read PM2.5 concentration

Send	FE A5 00 00 A5
Response	FE A5 02 00 DF1 DF2 [CS]
Description	PM2.5 value = DF1×256 + DF2 (unit: $\mu g/m^3$)

b)Read PM1.0, PM2.5, PM10 concentration

Table 9. Read PM1.0, PM2.5, PM10 concentration

Send	FE A5 00 01 A6
Response	FE A5 02 00 DF11 DF12 DF21 DF22 DF31 DF32 [CS]
Description	PM1.0 value=DF11×256+DF12(unit: $\mu g/m^3$) PM2.5 value=DF21×256+DF22(unit: $\mu g/m^3$) PM10 value=DF31×256+DF32(unit: $\mu g/m^3$)

c) Read PM1.0, PM2.5, PM10 concentration (reserved)

Table 10.Read PM1.0, PM2.5, PM10 concentration (reserved)

Send	FE A5 00 02 A7
Response	FE A5 02 00 DF11 DF12 DF21 DF22 DF31 DF32 DF41 DF42 DF51 DF52 [CS]

Description	PM2.5 value= DF11×256 + DF12(unit: $\mu g/m^3$) humidity = DF21×256+DF22(unit: %RH, *10 times, accuacy0.1) Temperature = DF31×256+DF32(unit: ℃, *10 times, accuracy 0.1) (Temperature value less than 500 is negative temperature, such as: 400 is -10°C, 600 is 10°C)

3.5 PWM output

Table 11. PWM output

PM2.5 measure range	0~1000μg/m^3
Period	1000ms±5%
High level output at the beginning of the cycle	200μs (theoretical value)
middle cycle	1000ms±5%
Low level output at the end of the period	200μs (theoretical value)

Note: 1) The calculation formula for obtaining the current PM2.5 concentration value through PWM:
P=1000(TH)/(TH+TL);
P is the PM2.5 concentration value, the unit is /;
TH is the high level time in one output cycle, and TL is the low level time in one output cycle.
2) The value calculated by PWM only indicates PM2.5; PWM output, active high.

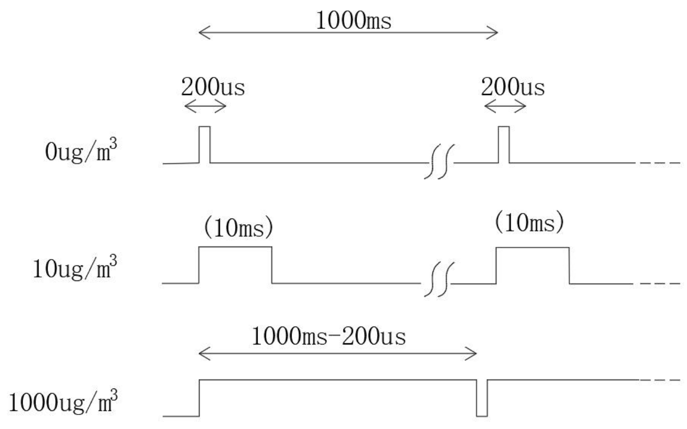


Figure 5. PWM output timing diagram

4 Dimension

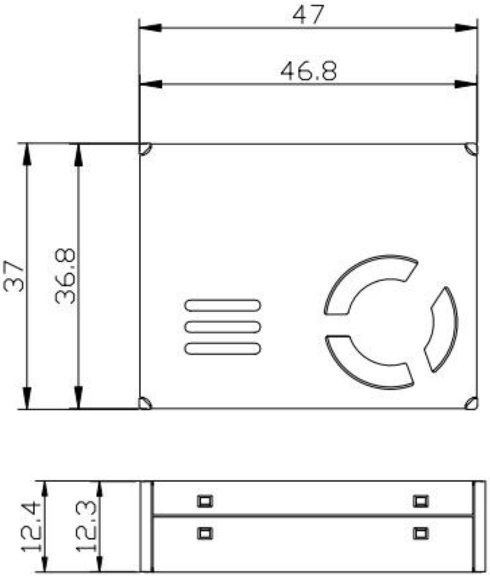


Figure 6. APM2000 dimension (unit:mm, tolerance: ISO2768-mK)

5 Packing

APM2000 use plastic tray for packing, each tray has 25pcs as show below:

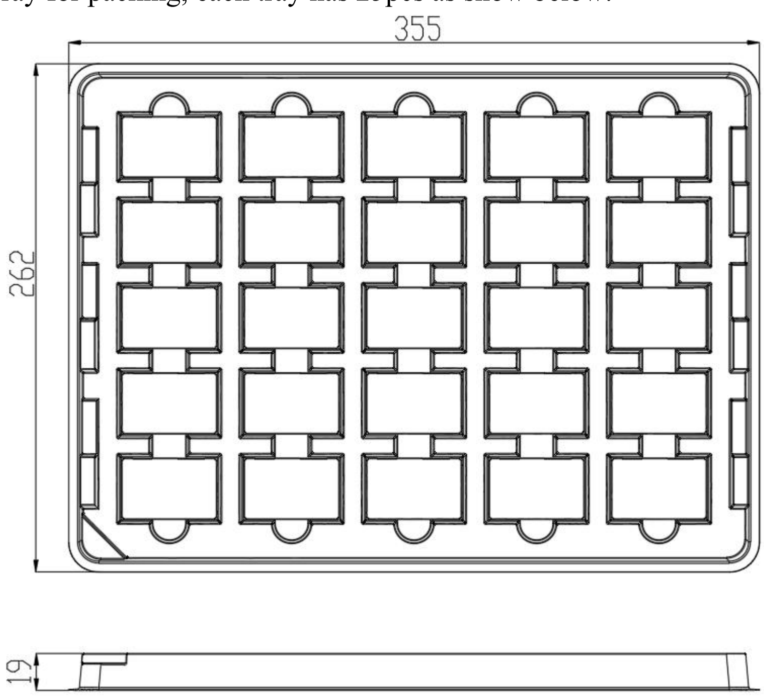


Figure 7: Plastic tray packing

Each sensor weighs around 26g; each tray weighs around 650g.

6 Notes

6.1 Since the metal shell of the sensor is connected to the internal power supply ground, please do not short-circuit the sensor shell with other external circuits or the chassis shell.

6.2 The best way to install the sensor is to place the plane where the air inlet and outlet are located close to the air holes connecting the inside and outside of the user's whole machine. There should be no shelter within 2cm around the air outlet. There should be an airflow isolation between the air inlet and the air outlet to prevent the air from returning directly from the air outlet to the air inlet inside the device.

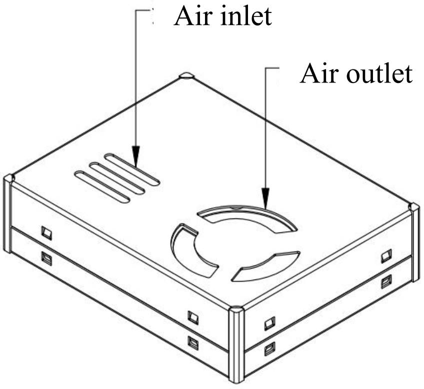


Figure 8. Schematic diagram of air inlet and outlet

6.3 The opening size of the air inlet and outlet of the equipment should not be smaller than the opening size of the air inlet and outlet of the sensor.

6.4 When used in purifier products, avoid placing the sensor directly in the air duct of the purifier itself. An independent structural space should be designed in which the sensor is placed to isolate the sensor from the air duct of the purifier itself.

6.5 The installation position of the sensor should be 20cm higher than the ground, otherwise there may be large dust particles or even flocs such as ground sand, floating flocs, etc., which may cause the fan to wind and block rotation, affecting the measurement. Proper pre-filtering is recommended for user equipment.

6.6 Users must not disassemble the sensor, including the metal shielding case, to prevent irreversible damage.

6.7 The data of the factory sensor has been tested and the data consistency is good. Do not use third-party testing instruments or data as a comparison standard. If the user wants the measurement data to be consistent with the third-party testing equipment, data fitting and calibration can be performed according to the actual measurement results.

6.8 This sensor is suitable for ordinary indoor environments. If the user equipment is used in the following environments, the data consistency of the sensor may decrease due to excessive dust accumulation, oil accumulation, and water ingress:

a) The dust concentration is greater than 300/ in 50% of the year, or greater than 500/ in 20% of the time

b) Fume environment

c) High water mist environment

d) outdoor

WARNINGS AND PERSONAL INJURY

Do not use this product in safety protection devices or emergency stop equipment, and in any other application where failure of the product may cause personal injury, unless there is a specific purpose or authorization for use. Refer to the product data sheet and instruction manual before installing, handling, using or maintaining this product. Failure to follow recommendations could result in death or serious personal injury. The company will not be responsible for all compensation for personal injury and death arising therefrom, and exempt any claims that may arise from the company's managers and employees, affiliated agents, distributors, etc., including: various costs, claims fees, attorney fees, etc.

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Product	Warranty
APM2000 PM sensor	12 month

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